

THE ASYMPTOTIC POWER OF A CLASS OF GOODNESS OF FIT TEST CRITERIA

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SUMMARY

This paper provides a method for the calculation of the asymptotic power, under alternatives close to the null hypothesis, of the class of goodness of fit test criteria Q_n^W , of which the asymptotic null distribution was studied in De Wet and Venter (1973). This is done by reducing the problem to that of calculating the asymptotic distribution of a quadratic form plus a linear term.

1. Introduction

Let $Z_1, Z_2, \dots$ be independent and identically distributed random variables with $EZ_1 = 0$, $\text{Var } Z_1 = 1$ and $EZ_1^4 < \infty$. Let $\{c_{ijn}; i, j = 1, \dots, n\}$ and $\{d_{in}; i = 1, \dots, n\}$ be real numbers with $c_{ijn} = c_{jin}$ for all i and j and put

$$(1) \quad T_n = \sum_{i,j=1}^n c_{ijn} Z_i Z_j + \sum_{i=1}^n d_{in} Z_i.$$

In section 2 we give some results on the characterization of the asymptotic distribution of T_n as $n \rightarrow \infty$ under suitable assumptions on the c_{ijn} 's and d_{in} 's. These results are closely related to those of section 2 of De Wet and Venter (1973).

Let $X_{1n}, X_{2n}, \dots, X_{nn}$ be the order statistics of an independent sample of size n from the distribution F and suppose $H_0: F = F_0$ is to be tested. Let W be a weight function on $(0,1)$; in De Wet and Venter (1973) we considered

$$(2) \quad Q_n^W = \sum_{k=1}^n (F_0(X_{kn}) - k/(n+1))^2 W(k/(n+1))$$

as a test statistic for H_0 . In section 3 we show how to calculate the asymptotic distribution and power of Q_n^W under alternatives of the form

$$H_1 : F^{-1} = F_n^{-1} = F_0^{-1} + (n+1)^{-1/2} G$$

for some function G .

In section 4 we illustrate with the example of testing for a $N(0,1)$ distribution with location and scale alternatives.

2. Main Results

In this section we give two theorems which characterize the asymptotic distribution of T_n . Since these theorems are closely related to theorems 1 and 2 of De Wet and Venter (1973), we will omit the proofs; details can be found in De Wet (1972).

Theorem 1.

Let $\{b_{imn}; i = 1, \dots, n; n, m = 1, 2, \dots\}$, $\{\gamma_m, m = 1, 2, \dots\}$

and

$$\{\delta_m, m = 1, 2, \dots\}$$

be real numbers (not all zero); let $\{M_n\}$ be a sequence of integers with $M_n \rightarrow \infty$, put

$$(3) \quad t_{1n} = \sum_{m=1}^{M_n} |\gamma_m|, t_{2n} = \sum_{m=1}^{M_n} |\delta_m|, t_n = \max(t_{1n}, t_{2n})$$

and assume that

$$(B1) \quad \max_{1 \leq i \leq n} |b_{imn}| \rightarrow 0 \text{ for each } m$$

$$(B2) \quad \max_{1 \leq m, k \leq M_n} \left| \sum_i b_{imn} b_{ikn} - \delta_{mk} \right| = o(t_n^{-1})$$

$$(B3) \quad \max_{1 \leq m, k \leq M_n} \sum_i b_{imn}^2 b_{ikn}^2 = o(t_{1n}^{-2})$$

$$(C) \quad \sum_{i,j} c_{ijn}^2 \rightarrow \sum_m \gamma_m^2 < \infty$$

$$(D) \quad \sum_i d_{in}^2 \rightarrow \sum_m \delta_m^2 < \infty$$

$$(BC) \max_{1 \leq m \leq M_n} \left| \sum_{i,j} c_{ijn} b_{imn} b_{jmn} - \gamma_m \right| = o(t_{1n}^{-1}).$$

$$(BD) \max_{1 \leq m \leq M_n} \left| \sum_i d_{in} b_{imn} - \delta_m \right| = o(t_{2n}^{-2}).$$

Then

$$(4) \quad D(T_n - ET_n) \rightarrow D(\sum_m \gamma_m (Y_m^2 - 1) + \sum_m \delta_m Y_m).$$

Theorem 2.

Let $\{b_{imn}\}$ satisfy

$$(B4) \quad \sum_i b_{imn} b_{ikn} \rightarrow \delta_{mk}$$

$$(B5) \quad n^{1/2} |b_{imn}| \leq b < \infty \text{ for all } i, m, n.$$

Let $\{\gamma_m\}$ and $\{\delta_m\}$ be real sequences for which (C) and (D) hold as well as

$$(C1) \quad \sum_m |\gamma_m| < \infty$$

$$(D1) \quad \sum_m |\delta_m| < \infty$$

$$(BC1) \quad \sum_{i,j} c_{ijn} b_{imn} b_{jmn} \rightarrow \gamma_m$$

$$(BD1) \quad \sum_i d_{in} b_{imn} \rightarrow \delta_m$$

Then (4) holds.

Remark.

One source of applications for these theorems is that in which

$$(5) \quad c_{ijn} \approx n^{-1} c(i/(n+1), j/(n+1)), d_{in} \approx n^{-1/2} d(i/(n+1))$$

with c a measurable square integrable function on the unit square symmetric in its arguments and d a measurable square integrable function on $(0,1)$.

Let $\{g_m\}$ and $\{\gamma_m\}$ be the eigenfunctions and -values of c and suppose $\{g_m\}$ is a

complete orthonormal system. Let $\{\delta_m\}$ be the Fourier coefficients of d with respect to $\{g_m\}$. If we then select

$$(6) \quad b_{imn} = n^{-1/2} g_m(i/(n+1))$$

we can often expect that the conditions of theorem 1 or 2 would apply.

3. Applications to Power Studies

Consider the test of fit problem mentioned in section 1 and assume that H_1 is true. In this section we indicate how the theory of the previous section can be applied to obtain the asymptotic distribution of Q_n^W . We will avoid writing down in detail the conditions under which the results hold for general F_0 , G and W since it becomes rather messy. Assume that F_0 is continuous and strictly increasing with first two derivatives f_0 and f_0' . Let $F_0^{-1} = H$ and for a function f on $(0,1)$ write $f_{kn} = f(k/(n+1))$. Applying Taylor's theorem, with $U_{kn} = F_n(X_{kn})$, we have

$$(7) \quad F_0(X_{kn}) = F_0(H(U_{kn}) + (n+1)^{-1/2} G(U_{kn})) \\ = U_{kn} + (n+1)^{-1/2} G(U_{kn})f_0(H(U_{kn})) + 1/2 (n+1)^{-1} G(U_{kn})^2 f_0'(Z_{kn})$$

with Z_{kn} between $H(U_{kn})$ and $H(U_{kn}) + (n+1)^{-1/2} G(U_{kn})$. Assuming G' to exist and applying Taylor's theorem again, we have

$$(8) \quad G(U_{kn}) = G_{kn} + V_{kn} G'(\alpha_{kn})$$

$$(9) \quad f_0(H(U_{kn})) = f_0(H_{kn}) + (H(U_{kn}) - H_{kn})f_0'(\beta_{kn})$$

with α_{kn} and β_{kn} defined in the usual way and $V_{kn} = U_{kn} - k/(n+1)$.

Substituting from (7), (8) and (9) and multiplying out we have

$$(10) \quad Q_n^W = \sum_k V_{kn}^2 W_{kn} + 2(n+1)^{-1/2} \sum_k V_{kn} W_{kn} G_{kn} f_0(H_{kn}) \\ + (n+1)^{-1} \sum_k G_{kn}^2 f_0'(H_{kn}) W_{kn} + R_{1n} \\ = Y_{1n} + Y_{2n} + A_n + R_{1n} \text{ (say)}$$

with R_{1n} the remainder terms.

In many cases we have $R_{1n} = o_p(1)$ and

$$(11) \quad A_n \rightarrow \int_0^1 G^2(x) f_0^2(H(x)) W(x) dx = A < \infty.$$

Consider Y_{1n} and Y_{2n} . As in De Wet and Venter (1973) we may write $U_{kn} = S_k/S_{n+1}$ where

$$S_k = Z_1^* + \dots + Z_k^*$$

with $\{Z_i^*\}$ independent random variables each exponentially distributed with expectation one. Putting $Z_i = Z_i^* - 1$ and

$$\psi_{ijn} = \psi(i/(n+1), j/(n+1)) \text{ with}$$

$$(12) \quad \psi(x, y) = \begin{cases} 1-y & \text{for } x \leq y \\ -y & \text{for } x > y \end{cases}$$

it then follows that

$$(13) \quad Y_{1n} = (n+1/S_{n+1})^2 \sum_{i,j} c_{ijn+1} Z_i Z_j$$

and

$$(14) \quad Y_{2n} = (n+1/S_{n+1}) \sum_i d_{in+1} Z_i$$

with

$$(15) \quad c_{ijn+1} = (n+1)^{-2} \sum_{k=1}^n W_{kn} \psi_{ikn} \psi_{jkn}$$

and

$$(16) \quad d_{in+1} = 2(n+1)^{-3/2} \sum_{k=1}^n W_{kn} \psi_{ikn} G_{kn} f_o(H_{kn}).$$

Using this we can write

$$(17) \quad Q_n^W = (n+1/S_{n+1})^2 T_{n+1} + A_n + R_n$$

with T_n as in (1) and

$$(18) \quad R_n = R_{1n} + (n+1/S_{n+1}) (1 - n+1/S_{n+1}) \sum_i d_{in+1} Z_i.$$

If $R_{1n} = o_p(1)$ we can often expect R_n to be $o_p(1)$ too, since in many cases $\sum_i d_{in+1} Z_i$ is asymptotically normally distributed. If this is the case, and (11) holds, (17) implies that Q_n^W should have the same asymptotic distribution as $T_n + A$.

Remarks.

1. Note that in this case $c_{ijn} \approx n^{-1} c_w(i/(n+1), j/(n+1))$ and $d_{in} \approx n^{-1/2} d_w(i/(n+1))$ with

$$(19) \quad c_w(x, y) = \int_0^1 W(z) \psi(x, z) \psi(y, z) dz$$

and

$$(20) \quad d_w(x) = 2 \int_0^1 W(z) \psi(x, z) G(z) f_0(H(z)) dz$$

assuming f_0 , G and W to be such that these integrals exist. Hence if c_w and d_w are also quadratically integrable we can expect to apply the theorems of the previous section in the way indicated in the remark above.

2. The characteristic function of the asymptotic distribution in (4) can be calculated and inverted numerically in the same way as in De Wet and Venter (1972). If all the above hold, this would then yield the asymptotic power of Q_n^W .

4. Examples

Let Φ and ϕ be the $N(0,1)$ distribution and density functions respectively and set

$$(21) \quad h(x) = 1/\phi(\Phi^{-1}(x))^2.$$

In De Wet and Venter (1972) we considered Q_n^h as a test statistic of the hypothesis $H_0: F_0 = \Phi$. In this section we illustrate the above theory by calculating the asymptotic power of Q_n^h for the simple cases of location and scale alternatives. (Of course if H_0 is to be tested against these alternatives, more powerful tests exist, but the present intention is merely to illustrate the above theory.)

(i) Location Alternatives

For $\theta \neq 0$, let $G(x) \equiv \theta$, i.e. $F_n(x) = \Phi(x - \theta(n+1)^{-1/2})$. The proof that $R_n = o_p(1)$ is quite lengthy and will not be given here; details can be found in De Wet (1972). Obviously $A_n = \theta^2 n/(n+1)$ so $A = \theta^2$. To obtain the asymptotic distribution of T_n we apply the remark above. In De Wet and Venter (1973) we showed that the eigenfunctions and -values of $c_h(x, y)$ are

$$(22) \quad g_m(x) = h_m(\Phi^{-1}(x)), \gamma_m = m^{-1}, m = 1, 2, \dots$$

with h_m the m -th normalized Hermitian polynomial. We also have

$$(23) \quad d_h(x) = 2\theta \int_0^1 \psi(x, z) dz / \phi(\Phi^{-1}(z))$$

getting

$$(24) \delta_m = \int_0^1 d_h(x) g_m(x) dx.$$

Substituting for d_h and changing the order of integration yield

$$\begin{aligned} \delta_m &= 2\theta \int_0^1 \phi(\Phi^{-1}(z))^{-1} \left(\int_0^1 \psi(x, z) g_m(x) dx \right) dz \\ &= -2\theta m^{-1/2} \int_0^1 g_{m-1}(z) dz. \end{aligned}$$

Using orthonormality of $\{g_m\}$ and the fact that $g_0 \equiv 1$ we have

$$(25) \delta_m = \begin{cases} 0 & \text{for } m > 1 \\ -2\theta & \text{for } m = 1. \end{cases}$$

With this choice of $\{\gamma_m\}$ and $\{\delta_m\}$ the conditions of theorem 1 hold (see De Wet 1972 for details) and we have

$$(26) D(T_n - ET_n) \rightarrow D(\sum_m^{-1} (Y_m^2 - 1) - 2\theta Y_1).$$

Let

$$(27) a_n^0 = ET_{n+1} = (n+1)^{-1} \sum_{k=1}^n k/(n+1) (1 - k/(n+1)) h_{kn}.$$

It follows from De Wet and Venter (1973) that $a_n^0 = o(\log n)$. This together with the fact that $n+1/S_{n+1} = 1 + O_p(n^{-1/2})$ give

$$(28) D(Q_n^h - a_n^0) \rightarrow D(\sum_{m=1}^{\infty} m^{-1} (Y_m^2 - 1) - 2\theta Y_1 + \theta^2).$$

The characteristic function of the asymptotic distribution in (28) can be inverted numerically as in De Wet and Venter (1972) and from this we get the asymptotic power, given in the table below for $\theta = 1$.

(ii) For $\theta > 0$ take $G(x) = \theta \Phi^{-1}(x)$ to get $F_n(x) = \Phi(x(1 + \theta(n+1)^{-1/2}))$. Proceeding as in (i) we get $\{\gamma_m\}$ as before while

$$(29) \delta_m = \begin{cases} 0 & \text{for } m \neq 2, m \geq 1 \\ -\sqrt{2}\theta & \text{for } m = 2 \end{cases}$$

and $A = \theta^2$.

Theorem 1 can be applied again (see De Wet (1972) for the verification of the

conditions) and we obtain

$$(30) \quad D(Q_n^h - a_n^0) \rightarrow D\left(\sum_{m=1}^{\infty} m^{-1} (Y_m^2 - 1) - \sqrt{2}\theta Y_2 + \theta^2\right).$$

Numerical inversion of the characteristic function of this asymptotic distribution can be done as before and we obtain the power, given below, for $\theta = 1$.

TABLE 1
Asymptotic power of Q_n^h for location and scale alternatives

Asymptotic Power		
Test size	Location Alternatives	Scale Alternatives
0,20	0,3622	0,3876
0,15	0,3033	0,3173
0,10	0,2359	0,2359
0,05	0,1527	0,1371
0,01	0,0538	0,0342

Remark

The above theory has also been applied to calculating the asymptotic power of the statistic for the composite hypothesis $H_0 : F_0(x) = \Phi((x - \mu)/\sigma)$, considered in De Wet and Venter (1972). Details of this can be found in De Wet (1972).

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